Sci-Verify: A Forensic SaaS for Scientific Information Retrieval and Citation Provenance

Bendourou Sami, N2503395D

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Code available in GitHub :[https://github.com/bend0u/Citation-Checker]

1 Introduction

LLM hallucinations are a major problem for researchers. When AI is used to find or cite sources, it often creates fake or irrelevant references, leading to a breakdown in scholarly trust known as the "provenance problem". To fix this, we developed Sci-Verify, a SaaS tool that uses Adversarial RAG technology. The platform allows users to either retrieve real citations that answer a specific question or validate if a claim actually exists within a cited paper. The following user stories explain how a researcher uses these features to find and prove scientific evidence.

2 User Stories: The Core Workflows

The design of Sci-Verify is driven by two primary user workflows tailored to the needs of modern researchers.

2.1 User Story 1: The Deep Research Query

As a researcher, I want to enter a complex scientific question and receive a synthesized, brief answer that proves its claims.

- **Brief Synthesis**: The system must provide a concise answer to the question based on peer-reviewed data.
- **Source Retrieval**: The tool must retrieve relevant papers directly related to the question, providing clickable links and direct DOI access for every source.
- **Citation List**: The system must generate a list of relevant, verified citations that I can immediately use in my own writing.

2.2 User Story 2: The Citation Verification (External Audit)

As an author who has already drafted text with an LLM, I want to verify specific citations and papers.

- **Upload & Entry**: I want to upload or paste a specific claim along with its cited paper.
- **Paper Existence Check**: The system must first query global registries like OpenAlex to verify if the cited paper actually exists, alerting me to "shadow citations".
- **Internal Evidence Verification**: If the paper exists, the system must retrieve it and verify if the specific claim or quote can be found within the text of that paper.
- **Provenance Trace**: If verified, I want to see the exact paragraph from the original paper that supports the claim.

3 Literature Review

The current landscape of AI-assisted research is defined by a tension between high-speed synthesis and the preservation of scientific integrity. This review examines the theoretical challenges and emerging architectural frameworks that inform our project's focus on deep search and citation verification.

3.1 The Provenance Problem in Scholarly Writing

The fundamental challenge in AI-driven research is what Earp et al. (2025) [1] define as the "provenance problem." In their analysis, *"LLM use in scholarly writing poses a provenance problem"*, they argue that large language models systematically break the chain of scholarly acknowledgment. Because these models synthesize information without a deterministic link to the original human authors, they frequently produce "shadow citations", references that appear authentic but lead to 404 errors or irrelevant sources. This research highlights the urgent need for a verification system that can re-establish the link between a claim and its actual existence within a cited document.

3.2 Standardizing the Audit: The DeepTRACE Framework

To address the unreliability of generative tools, Venkit et al. (2026) [2] proposed the **DeepTRACE** framework. Their study conducted a forensic audit of popular deep research agents and found that citation accuracy—the rate at which a cited source actually supports the claim—ranges from only 40% to 80%. This research is relevant to the development of "Deep Search" features, as it provides the metrics required to measure success, such as the *Factual Support Matrix* and *Citation Thoroughness*. Their findings suggest that verbosity in AI responses often creates a "false impression of validation," making it necessary for search systems to prioritize brief, verified answers over long, unsupported summaries.

3.3 Architectural Trends: Adversarial Collaboration (AC-RAG)

While the provenance problem defines the "why" and DeepTRACE defines the "what," recent architectural research suggests a specific "how" for solving retrieval failures. Zhang et al. (2025) [3] introduced the concept of **Adversarial Collaboration RAG (AC-RAG)** in their paper, *"Enhancing Retrieval Augmentation via Adversarial Collaboration"*. Although the authors temporarily withdrew the paper for further review in early

2026, the framework remains highly relevant. It proposes that "Retrieval Hallucinations" can be mitigated by pitting a "Detector" agent against a "Resolver" agent.

This adversarial approach suggests that a system designed to retrieve and verify papers should not simply trust its first synthesis. Instead, the most robust results are achieved when an internal auditor node is used to challenge and verify every claim against a deterministic registry like OpenAlex before the result is presented to the user.

4 System Architecture: Deep Search (Discovery Pipeline)

The Deep Search engine is designed to handle research queries, such as *"What are the three main properties of the Raft consensus algorithm as defined by Ongaro and Ousterhout"*, by transforming abstract questions into a synthesized answer backed by literal evidence. To avoid the "provenance problem" identified in current AI systems, this architecture prioritizes the retrieval of relevant papers before any text generation occurs.

4.1 Stage 1: Multi-Level Query Deconstruction

The pipeline begins by parsing the user's raw input to isolate hard search parameters.

- **Keyword Tiering**: The system decomposes the query into a prioritized list of keywords, moving from highly specific (e.g., "Raft consensus") to more general domain terms, like "Consensus Algorithm".
- **Author Filtering**: It identifies specific researchers mentioned in the query (e.g., "Diego Ongaro") to create a strict "Keyword + Author" search constraint.
- **Why**: This ensures the system does not just search for a general topic, but actively hunts for the specific primary sources required to answer the question.

4.2 Stage 2: Abstract Relevance Scoring (The Judge)

Because initial searches can return high "noise" or false positives, the architecture employs a rigorous filtering stage.

- **Relevance Assessment**: A "Judge" model evaluates all retrieved abstracts against the original user query, assigning each a score between 1 and 100.
- **Strict Thresholding**: Only papers scoring ¿= 50 are retained. The system then selects the top three most relevant papers to proceed to full-text analysis.
- **Why**: This adversarial filtering prevents the final synthesis from being polluted by irrelevant data, ensuring the "Executive Summary" is based only on the most pertinent scientific evidence.

4.3 Stage 3: Full-Text Extraction and Semantic Indexing

Once the top sources are selected, the system shifts from metadata to deep content analysis.

- **PDF Parsing and Chunking**: The system automatically extracts the raw text from the papers and breaks them into overlapping "textual windows" (500 characters).
- **Vector Indexing**: These chunks are converted into dense mathematical vectors and stored in a memory-resident database (FAISS).
- **Why**: This allows the system to perform granular retrieval, finding the exact paragraph or sentence that directly answers the user's question, rather than summarizing the entire paper at once.

4.4 Stage 4: Synthesis and Verbatim Quote Extraction

The final stage uses the most relevant "evidence snippets" found in the vector index to generate a final report.

- **Executive Summary**: The system provides a brief, synthesized introduction that directly addresses the question (e.g., listing the 3 properties of Raft).
- **Exact Citations**: Crucially, the system extracts literal, verbatim strings directly from the source material. These quotes are provided in a format that the researcher can immediately copy and paste into their own work.
- **Why**: By forcing the model to provide verbatim quotes, the architecture ensures forensic-level accuracy. Any attempt by the AI to paraphrase or "hallucinate" evidence is blocked by strict structural constraints.

5 System Architecture: Verifier of Citations

While the Deep Search focuses on the discovery of new information, the Verifier of Citations is a dedicated pipeline designed to audit text that has already been generated. Its purpose is to solve the "provenance problem" by determining if the claims made by an AI are actually supported by the papers it cites.

5.1 Stage 1: Metadata Extraction

The process begins by extracting *only* sentences that make factual claims tied to academic research. These claims may be explicit quotes (demarcated by quotation marks) or semantic assertions where a specific topic is attributed to a paper.

- **Data Collection**: The system identifies the specific segments of a sentence that claim factual status and extracts the "claimed" metadata: the paper title, the authors, and the publication year.
- **Why**: AI-generated research often buries facts in long, conversational paragraphs. To verify them, we must first translate this prose into a structured format suitable for precise registry searching.

5.2 Stage 2: Flexible Metadata Reconciliation (The Search Waterfall)

The most critical phase of the audit is locating the actual paper in a global registry, even if the AI provided slightly incorrect or abbreviated information. This is achieved through a process of **Progressive Relaxation Search**.

- **The Problem of Imperfect Data**: Imagine an AI claims a fact originates from the paper *"Attention is All You Need"* by *Vaswani* in **2018**. A strict registry search might fail because the paper was officially published in **2017**.
- **The Solution (Relaxation)**: The system is designed to be flexible. If a strict search fails, it "relaxes" the search constraints—dropping the year to search by title/author or using fuzzy matching to find a paper even if the title is abbreviated (e.g., locating the full record for *"The Raft Paper"*).
- **Why**: DeepTRACE findings demonstrate that AI agents frequently hallucinate or misattribute parts of a citation. Without this flexibility, the system would falsely report that a paper "does not exist" due to a minor typographical error.

5.3 Stage 3: Internal Evidence Retrieval

Once the correct paper is identified and retrieved, the system must check if the specific claim actually exists within that document's content.

- **Contextual Indexing**: The system reads the paper's content and creates a searchable semantic map of its paragraphs.
- **Snippet Matching**: It uses the original claim as a mathematical query to find the most relevant sections of the paper that should contain the proof.
- **Why**: Simply finding the source paper is insufficient; we must ensure the document provides factual support for the specific claim made by the user.

5.4 Stage 4: Final Verification Verdicts

The final stage of the citation verification issues a definitive verdict by routing the claim through two distinct logic pathways based on how the information was originally framed.

- **Path A: Deterministic String Matching (For Literal Quotes)**: If a claim is identified as an explicit, word-for-word quote, the system bypasses generative models to avoid the risk of secondary hallucinations. Instead, it uses a mathematical similarity check to compare the user's quote against the retrieved text from the actual paper. A match of $\geq 85\%$ is required to issue a **Verified Exact Quote** status.
- **Path B: Semantic Inference Judge (For Summaries)**: For claims written as general summaries, the system employs an "Inference Judge". This model evaluates "Entailment": it determines if the retrieved evidence from the paper contains sufficient semantic proof to make the user's claim factually true.

- **Handling Inaccessible Data**: In cases where a paper is paywalled or the full text cannot be retrieved, the system "fails safely" by flagging the result as **Unknown/-Paywalled** rather than attempting to guess.
- **Why**: This dual-pathway approach is designed to eliminate the failure modes identified in the DeepTRACE study, where systems often present "highly confident" but "unsupported" statements. By separating verbatim quotes from semantic summaries, Sci-Verify ensures that "Hallucinated Quotes" are caught with mathematical certainty, while "Unsupported Claims" are filtered through rigorous reasoning.

6 Technical Implementation

The implementation of Sci-Verify focuses on balancing high-speed execution with the rigorous citation standards defined in the DeepTRACE framework. We utilize a Python-based ecosystem to orchestrate several specialized services and models.

6.1 Python and Streamlit

The core application is developed in **Python**, chosen for its mature ecosystem in natural language processing and agentic orchestration. The user interface is built using **Streamlit**, which allows for the rapid deployment of a clean, responsive dashboard where researchers can interact with the Deep Search and Citation Verifier without local setup.

6.2 Scholarly Registry: OpenAlex

To solve the "provenance problem," we use **OpenAlex** as our primary source of truth.

- **Access**: We utilize the `pyalex` library to interact with the OpenAlex API, which provides access to over 250 million scholarly records.
- **Data Quality**: Unlike standard web-search engines that return high noise, OpenAlex provides structured metadata (DOI, authors, and concepts), which is essential for the "Metadata Extraction" stage of our verifier.

6.3 Inference Engine: Groq LPU

A significant challenge in multi-agent RAG systems is the "latency tax", the cumulative time required to complete multiple sequential and parallel LLM calls.

- **Performance**: We utilize the **Groq LPU** (Language Processing Unit) architecture to access various Large Language Models. This platform provides the sub-second inference speeds necessary to run our complex adversarial loops in real-time without degrading the user experience.
- **High-Frequency Logic**: Because our architecture requires multiple discrete calls for query deconstruction, relevance judging, and NLI verification, the high-throughput nature of Groq is a critical enabler for the system's "Deep Search" functionality.

- **Course Scope and Accessibility**: For the requirements of this project, the Groq Free Tier is sufficient. It provides robust access to a variety of high-performing open-source models, such as the Llama-3 series, which are ideal for academic and prototypical use cases. For more advanced tasks and more funding we could use better models.

6.4 Model Specialization and Tiering

We employ a tiered model strategy to maximize both speed and reasoning quality:

- **Llama-3.1-8B**: Used for "narrow" tasks such as query deconstruction and initial metadata extraction. These tasks require rapid pattern matching rather than deep reasoning.
- **Llama-3.3-70B**: Reserved for the final "Synthesis" stage. Its higher parameter count is necessary to ensure the executive summary is grammatically sophisticated and adheres to strict factual constraints.
- **All-MiniLM-L6-v2**: A local embedding model used to power the FAISS vector index, allowing for mathematically precise similarity searches between claims and paper segments.

6.5 Potential Challenges related to our choices

While this technology stack provides a robust solution, it involves several inherent trade-offs:

- **Paywall Limitations**: Even though OpenAlex identifies millions of papers, a considerable proportion is behind paywalls or suffer from 404 errors. In these cases, as a fallback, our system safely will analyze the abstract to avoid "guessing".
- **Rate Limits and Cost**: Running high-frequency LLM calls for every search query increases the "cost per search" compared to traditional keyword search. We mitigate this by using smaller models for non-critical tasks.

7 Experimentation and Qualitative Results

To evaluate the discovery pipeline, we conducted a test query regarding the Raft consensus algorithm. This experiment assesses whether the adversarial architecture can successfully provide grounded answers for technical queries where standard generative search engines often fail to provide factual support[cite: 23].

7.1 Deep Search Discovery

The first experiment involved inputting a specific scientific query into the discovery interface: "What are the three main properties of the Raft consensus algorithm as defined by Ongaro and Ousterhout" (see Figure 1).

The system generated an executive summary identifying leader election, log replication, and safety as the three core properties. As shown in Figure 2, the output includes

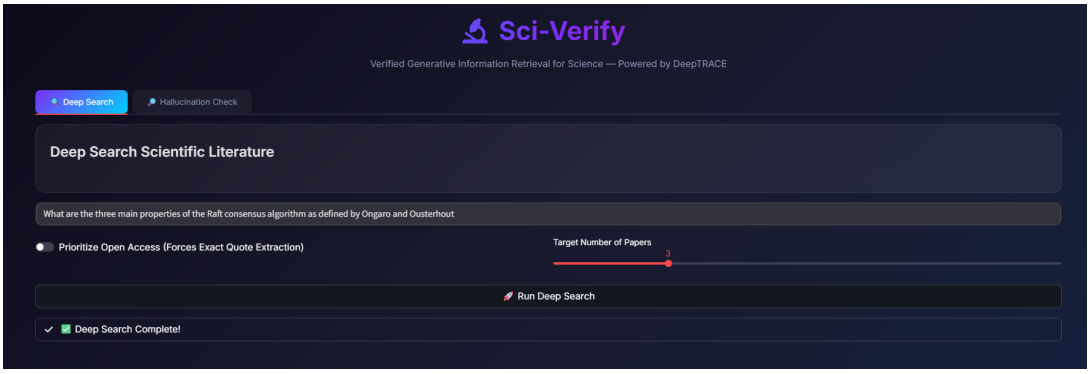

Figure 1: Deep Search query interface displaying the specific query for Raft consensus properties.

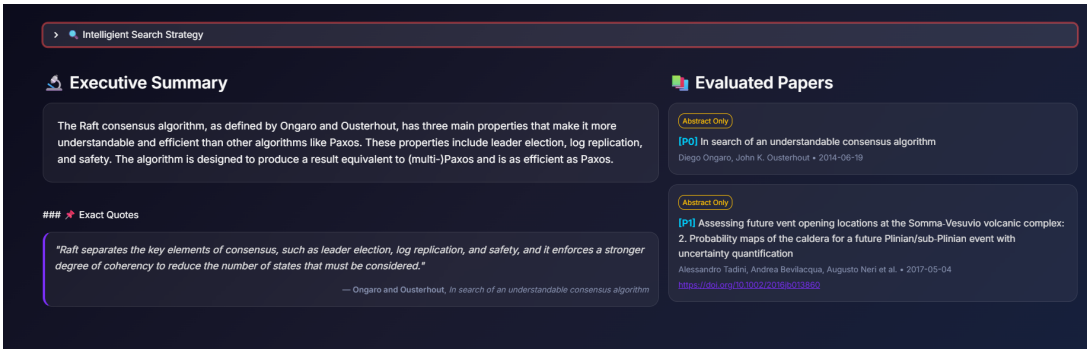

Figure 2: Discovery results displaying the synthesized summary, a verbatim quote with source metadata, and the list of evaluated papers.

an "Exact Quotes" section featuring a verbatim excerpt from the source paper, "In search of an understandable consensus algorithm," by Ongaro and Ousterhout.

The right-hand pane of the results screen displays the list of papers retrieved and evaluated by the relevance judge. For each paper, the system provides the title, author list, and publication date, alongside DOI links for direct verification in case it is Open Access.

7.2 Verifier of Citations

The second experiment evaluated the "Hallucination Check" pipeline, which serves as a forensic audit for existing text. This test fulfills User Story 2 by verifying whether a specific claim and its associated citation are factually grounded in the original source material.

As shown in Figure 3, the user provided the synthesized text from the discovery phase for verification. The system processed the input to extract the claimed title, author, and the specific verbatim string for auditing.

The resulting "Forensic Audit Report" (Figure 4) issued a **VERIFIED_QUOTE** status. The system achieved a 100% Verification Accuracy score based on the following automated reasoning:

- **Fuzzy Sequence Match**: Achieved a 100% similarity score by mathematically comparing the claim against the source text.

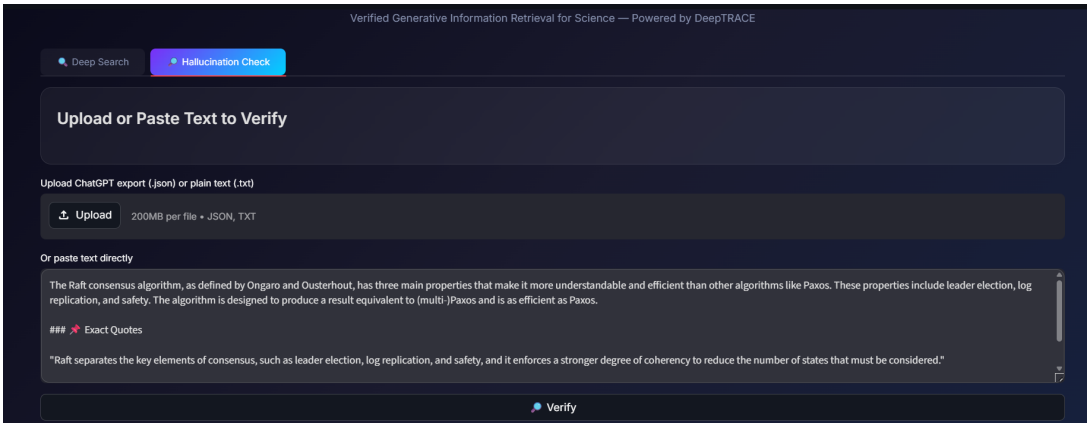

Figure 3: The Hallucination Check interface where the previously generated Raft summary and quote were pasted for verification.

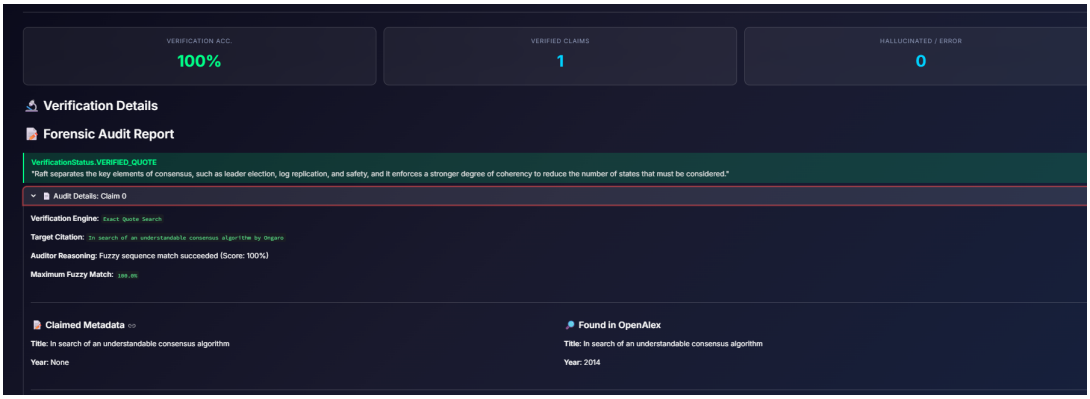

Figure 4: Forensic Audit Report showing 100% verification accuracy and successful metadata reconciliation.

- **Metadata Reconciliation**: Correctly identified the 2014 publication year via OpenAlex, resolving the missing year in the input.
- **Audit Verdict**: Confirmed the quote exists verbatim in the document, formally verifying the claim's provenance.

This experiment demonstrates that the adversarial auditing logic can effectively catch "shadow citations" and verify literal evidence with high precision, providing a transparent trust score for academic writing.

8 Discussion of Limitations

While Sci-Verify addresses the primary failure modes of modern research assistants, the system remains subject to several technical and data-driven limitations. These factors should be considered when evaluating the reliability of the output.

8.1 Open Access and Paywall Bottlenecks

The system's ability to perform deep verification is strictly bounded by the availability of full-text documents. Because the current implementation relies on open-access URLs

provided by OpenAlex, a big proportion of retrieved papers may be behind paywalls. for example, in 2024, only 23% of the papers were open access. In these instances, the system is forced to rely on abstracts only, which can lead to "Unknown" verdicts for claims that may actually exist within the hidden full-text content.

8.2 Keyword Sensitivity in Discovery

The effectiveness of the Deep Search is highly dependent on the initial query deconstruction phase. If the LLM "Reformulator" selects poor keywords or misses critical author names, the downstream retrieval engine may return irrelevant papers. This "garbage in, garbage out" risk means that the system is only as effective as the search parameters it generates at the start of the pipeline.

8.3 Limits of the Hallucination Rescue

The "Progressive Relaxation Search" used in the citation verifier is designed to fix minor AI hallucinations (such as incorrect dates or abbreviated titles). However, there is a technical limit to this flexibility. If an LLM hallucination is extreme—such as a completely fabricated title combined with an incorrect author—the system will be unable to locate a matching record in the registry and will correctly flag the source as a hallucinated paper.

8.4 Probabilistic Nature of LLM Logic

Despite the addition of adversarial auditing layers, the system relies on probabilistic models for relevance scoring and semantic inference (NLI).

- **Semantic Reasoning**: The NLI judge may occasionally misinterpret complex scientific nuances, leading to incorrect entailment verdicts.
- **Deterministic Thresholds**: The 85% similarity threshold for literal quotes is a fixed mathematical constraint. While this prevents hallucinations, it may trigger false negatives for older PDFs with poor OCR (Optical Character Recognition) where the text contains scanning artifacts.

Ultimately, Sci-Verify is a tool designed to assist human auditing rather than a perfect replacement for it. The probabilistic nature of LLMs means that 100% accuracy is never guaranteed, particularly in highly specialized or paywalled scientific domains.

9 Conclusion and Future Work

Sci-Verify successfully demonstrates that the "provenance problem" in scholarly research can be mitigated through a specialized Generator-Auditor architecture that prioritizes verbatim evidence over synthetic summaries. The system effectively identifies "shadow citations" and resolves metadata errors through progressive relaxation search, although its overall utility is currently limited by paywall-driven data gaps, sensitivity to initial keyword selection, and the probabilistic reasoning ceiling of free-tier models. To address these bottlenecks, future development should focus on integrating with institutional journal APIs for full-text access, utilizing web-scraping agents to broaden search surfaces, and transitioning to higher-parameter models to improve semantic entailment. Ultimately,

Sci-Verify provides a robust framework for more ethical AI-assisted research, ensuring that automated insights remain firmly anchored in provable human knowledge.

References

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